



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Ms. Mohammed Razia Alangir Banu
Student Name	Thota Jhanasri
Roll Number	2520090121
Branch	CS-IT
Course Outcome	CO-4

Case Study Topic: Tourist Package Ranking and Travel Itinerary Optimization using Heap Sort, Greedy and Dynamic Programming

Work Done Summary:

In this case study, advanced sorting and optimization techniques were implemented for the TourPlan – Smart Tourism & Travel Itinerary Optimization System. Tourist packages were organized and ranked using Heap Sort, Counting Sort, and Radix Sort based on popularity, ratings, and travel costs. Greedy Algorithm (Knapsack) was applied to select the best travel packages within a limited budget, while Dynamic Programming techniques such as 0/1 Knapsack, Longest Common Subsequence (LCS), and Matrix Chain Multiplication (MCM) were used to optimize itinerary planning and travel recommendations. The implementation demonstrates efficient sorting, optimal decision-making, and itinerary optimization by considering time complexity, space usage, stability, and problem characteristics.

Implementation Details:

1. Created a dataset of tourist packages containing package cost, popularity score, and customer ratings.
2. Implemented Heap Sort to rank tourist destinations based on popularity and travel demand.
3. Applied Counting Sort to efficiently sort destination ratings within a fixed range.
4. Implemented Radix Sort to organize travel package prices and budgets.
5. Used the Greedy Knapsack Algorithm to select the maximum number of attractions within a tourist's budget.
6. Implemented Longest Common Subsequence (LCS) to compare tourist preferences and recommend suitable
7. Applied Matrix Chain Multiplication (MCM) to optimize travel cost calculations and itinerary planning.
8. Evaluated algorithm performance based on time complexity, space usage, stability, and suitability for tourism optimization applications.

RESULT : SCREENSHOTS/OUTPUT

```
=== TOURPLAN PACKAGE RANKING SYSTEM ===

Popularity Scores Before Sorting:
[95, 88, 75, 90, 85]

Popularity Scores After Heap Sort:
[75, 88, 85, 90, 95]

Selected Tourist Packages:
Araku Cost = ₹1500
Vizag Cost = ₹2500
Warangal Cost = ₹1800

LCS Length = 5

C04 Execution Completed Successfully
```

The TourPlan system successfully ranked tourist packages using Heap Sort and optimized package selection using Greedy and Dynamic Programming techniques. Tourist preferences were analyzed using LCS to improve itinerary recommendations. The results demonstrate efficient travel planning, package ranking, and budget optimization for smart tourism applications.

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

Student Signature	Faculty Signature



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Course Outcome	CO-5

Case Study Topic: Tourist Route Recommendation and Travel Package Optimization using Non-Linear Data Structures, Greedy and Dynamic Programming Algorithms

Work Done Summary:

In this case study, practical applications of Non-Linear Data Structures, Greedy Algorithms, and Dynamic Programming techniques were developed for the TourPlan - Smart Tourism & Travel Itinerary Optimization System. Tourist destinations and travel routes were represented using graph structures to model connectivity between cities. Greedy algorithms were used to select cost-effective travel packages, while Dynamic Programming techniques were applied to optimize itinerary planning and resource allocation. The implementation demonstrates efficient route recommendation, package optimization, and travel decision-making for smart tourism applications.

Implementation Details:

1. Represented tourist destinations and travel routes using graph-based non-linear data structures.
2. Created weighted connections between cities based on travel distance and cost.
3. Implemented graph traversal techniques to explore connected tourist destinations.
4. Applied Greedy Algorithms to select optimal travel packages within budget constraints.
5. Used Dynamic Programming techniques to optimize itinerary planning and package selection.
6. Calculated minimum travel costs and efficient route recommendations.
7. Evaluated different travel plans based on budget, distance, and tourist preferences.
8. Verified the effectiveness of non-linear data structures, Greedy, and Dynamic Programming techniques in tourism optimization.

RESULT : SCREENSHOTS/OUTPUT

```
==== TOURPLAN ROUTE RECOMMENDATION SYSTEM ====

Tourist Route Traversal:
Hyderabad Warangal Vizag Araku Vijayawada

Selected Travel Packages:
Araku -> ₹1500
Vizag -> ₹2500
Warangal -> ₹1000

Optimized Itinerary Days = 7

C05 Execution Completed Successfully
```

The TourPlan system successfully recommended tourist routes using graph-based non-linear data structures and optimized travel package selection using Greedy Algorithms. Dynamic Programming techniques improved itinerary planning by maximizing resource utilization within given constraints. The results demonstrate efficient route planning, budget optimization, and smart travel decision-making for tourism applications.

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

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Course Outcome	CO-6

Case Study Topic: TourPlan – Smart Tourism & Travel Itinerary Optimization System using Advanced Non-Linear Data Structures and Application Development

Work Done Summary:

In this case study, advanced non-linear data structures were utilized to develop a smart tourism application called **TourPlan**. The system was designed to manage tourist destinations, travel routes, hotel information, and itinerary planning efficiently. Graphs were used to represent city connectivity, trees were used for destination categorization, and priority-based structures were applied for route recommendations. The implementation demonstrates how non-linear data structures can be used in real-world software applications for tourism management and travel optimization.

Implementation Details:

1. Developed the TourPlan application using non-linear data structures.
2. Represented tourist destinations and city connections using Graphs.
3. Created weighted edges based on travel distance, time, and cost
4. Used Tree structures to organize tourist destinations by category and region.
5. Implemented route recommendation features using graph traversal techniques.
6. Applied shortest-path concepts for efficient travel planning.
7. Managed destination information, hotels, and packages through structured data organization.
8. Generated optimized travel itineraries based on user preferences and budget constraints.

RESULT : SCREENSHOTS/OUTPUT

```
Sample Output

===== TOURPLAN SMART TOURISM SYSTEM =====

Available Tourist Destinations:
1. Hyderabad
2. Marangal
3. Vizag
4. Araku
5. Vijayawada

Recommended Tourist Route (DFS Traversal):
Hyderabad Marangal Vizag Araku Vijayawada

Travel Cost : Rs.4500
Travel Duration : 7 Hours

Best Hotels:
1. Hotel Green Valley
2. Tourist Residency

Optimized Itinerary Generated Successfully
CO6 Execution Completed Successfully
```

The TourPlan – Smart Tourism & Travel Itinerary Optimization System successfully managed tourist destinations, travel routes, hotel information, and itinerary planning using non-linear data structures. The application efficiently recommended travel paths, optimized route selection, and organized tourism data for better user experience. The results demonstrate the practical implementation of non-linear data structures in developing intelligent tourism and travel management applications.

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

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